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Open steel design and graphics

Software Testing Report

Short-span steel girder module in OsdagBridge

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Build tested	Windows 11, conda env osdagbridge-env, branch dev

Contents

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1. What I tested and why

OsdagBridge has a module for short-span steel girder (plate girder) highway bridges. The screening task asked for more than a single happy-path click of Design. I had to change basic geometry, additional section data, and custom steel and deck properties; run at least three designs after Unlock; check 3D CAD and plots; and compare what I typed with the internal input and output dictionaries.

I cloned the project, created the conda environment, and ran the desktop app (python -m osdagbridge.desktop). I also wrote a runner that calls the same backend the GUI uses: PlateGirderBridge.set_input, design, get_results, and reset (that last call is what Unlock does). Failed cases were left as failures. I did not patch the product to make the log look clean.

This PDF is the write-up of that work. The numbered tables and the screenshots are from the executed test suite (see evidence/logs/test_overview.json). Full side-by-side dictionary files are in test_cases/DICTIONARY_COMPARISON.md and evidence/dictionaries/.

2. How this maps to the screening task

The table below is the checklist I used. Every row is something the task asked for, and the right column is where the evidence sits in this report or in the ZIP.

Task asked for	What I did
Basic geometry (span, width, skew, median, footpath)	TC01 and TC08 at 30 m; TC03 with 12 deg skew, both footpaths and a median; TC04 at 20 m; TC05 at 45 m. Section 5.
Additional inputs (deck, wearing course, studs, factors)	TC06: deck 280 mm, wearing 22 kN/m3 x 80 mm, covers 40/45, stud 22 x 125, gamma_m0 = 1.15. Sections 5-7.
Custom steel and custom deck, not only defaults	TC02, TC07, TC09. Dictionaries keep the numbers. Analysis does not always use density. Sections 6 and 10.
At least 3 consecutive designs after Unlock	TC_MULTI_01: 25 m, then 30 m, then 35 m after reset(). Same sequence on the live window and in the video. Sections 8-9.
3D CAD and structural plots	Live OCC viewer + Plots dock (Section 9), exported OCC PNGs under evidence/screenshots/cad_3d/, and the video.
UI vs input dictionary vs output dictionary	Section 6 tables. Full dumps in DICTIONARY_COMPARISON.md.

Task asked for	What I did
Steel components and the deck	util.* , steeldesign.details.* , deck.report.* from TC02. Section 7.
Pass and fail log with exact parameters	Sections 4-5. Two E 350A failures are kept on purpose.
Code-level GitHub issues	Seven live issues on Nidhikhare12/OsdagBridge: #19, #20, #41-#45. Links in Section 10 and issues/ISSUE_LINKS.md.
Silent video of the workflow	OsdagBridge_Testing_Demo.mp4. Section 11.

3. How I ran the tests

Each automated case does the same three steps the GUI does. First set_input writes the planned values onto the backend. Then design() runs analysis and member design. Then get_results() is the output dictionary that fills the Output dock and the design report.

After a successful design I also asked for CAD through get_3d_cad_parameters (the OCC 3D view is built from that object) and I built the structural plots with the same matplotlib helpers the Plots dock uses. For the three-run case I called reset() between designs, which is the function Unlock calls from input_dock.

The screenshots in Section 9 are not drawings I generated for the report. They are window captures of the OsdagBridge desktop, maximised, with Basic Inputs on the left and CAD or plots on the right. I used those same frames for the silent video.

4. Result of every case

I ran 11 scenarios. **9 passed** (including pass-with-warnings) and **2 failed**.

The two hard failures are TC01 and TC04. Both use the default database grade E 350A. Design stops because Gs (shear modulus) is not on input_dict. That is a product bug, not a bad test. TC08 is the same 30 m geometry after I wrote Gs and the deck fck / fctm / Ecm keys the material dialog would have written. TC08 finishes, which is how I knew the missing Gs was the trigger.

Several passing cases still carry warnings. Those warnings are the defects in Section 10: custom density stored but not used, fatigue utilisation above 100% while design() still returns CAD, cross-bracing and end-diaphragm grades ignored, and missing DCR categories printed as 0.0.

Case	Result	What I saw
TC01_basic_optimized	FAIL	Default E 350A / M40, 30 m. Design stops: Gs is not set.
TC02_custom_materials	PASS_WITH_WARNINGS	Custom steel density 80 and custom deck. CAD and plots come out. Analysis still uses 78500 N/m3.
TC03_skew_footpath_median	PASS_WITH_WARNINGS	Skew 12 deg, both footpaths, median. Several util keys show 0.0 even though LTB is 20.44%.
TC04_span_min_boundary	FAIL	20 m, default E 350A. Same Gs stop as TC01.
TC05_span_max_boundary	PASS_WITH_WARNINGS	45 m custom materials. CAD produced. Fatigue UR 160%.
TC06_additional_inputs	PASS_WITH_WARNINGS	Non-default deck, wearing course, studs, gamma. Values reach CAD.
TC07_distinct_cb_ed_materials	PASS_WITH_WARNINGS	Three different custom steels stored. Design uses girder steel only.
TC08_db_grade_with_gs	PASS_WITH_WARNINGS	Same as TC01 but Gs and deck fck primed. Design completes.
TC09_custom_section_stiffeners	PASS_WITH_WARNINGS	Design Type = Custom. Explicit plates and two bearing stiffeners.
TC10_osi_roundtrip	PASS_WITH_WARNINGS	OSI reload into a new backend. Keys survive at API. GUI map is still wrong.

Case	Result	What I saw
TC_MULTI_01	PASS	Three designs after reset. CAD 25 m, then 30 m, then 35 m. No leftover span.

PASS_WITH_WARNINGS means design() returned and CAD was built, but I recorded a defect in the same run. I counted those as passes for the suite score and as bugs for the issue drafts.

5. Exact parameters I used

Unless a row says Custom, Design Type is Optimized. Project location is Mumbai for every case. That is the combination I saved in each .osi file under osi_files/.

Case	Geometry	Materials and extras
TC01	L=30 m, CW=7.5 m, skew=0, no median, no footpath	Database E 350A, M40
TC02	L=30 m, CW=7.5 m	custom_steel_350_490, density 80; custom deck density 26
TC03	L=28 m, CW=10 m, skew=12 deg, footpath Both, median Yes	custom_steel_300_440
TC04	L=20 m, CW=6 m	Database E 350A, M40
TC05	L=45 m, CW=12 m	Custom steel/deck, density 78.5 / 25
TC06	L=30 m, CW=7.5 m	Deck 280 mm; wearing 22 x 80; stud 22 x 125; gamma_m0=1.15; ecc=1.5 m
TC07	L=30 m, CW=7.5 m	Girder 350/80; cross bracing 250/70; end diaphragm 280/75
TC08	Same geometry as TC01	E 350A plus Gs and deck fck/fctm/Ecm written on input_dict
TC09	L=30 m; Design Type = Custom	Plates 1400x450x20 / 520x25, tw=12; bearing stiffeners=2
TC10	L=30 m	TC02 OSI loaded into a fresh backend
MULTI	25/7.5/0 deg then 30/8.5/5 deg then 35/9.5/10 deg	Custom materials; reset() between runs

6. UI values vs input dictionary vs output dictionary

After set_input I compared every planned field with input_dict. After design() I compared the same field with output_dict where that key exists. Match = Yes means the three sides agree within 1e-6. I print four tables here so the report can be read without opening JSON. The rest of the cases are in test_cases/DICTIONARY_COMPARISON.md.

Table 6.1 TC02 - custom steel and deck (stored correctly)

Key	I entered	Input dict	Output dict	Match
`geometry.span`	30.0	30.0	30.0	Yes
`geometry.carriageway_width`	7.5	7.5	7.5	Yes
`geometry.skew_angle`	0.0	0.0	0.0	Yes
`material.girder`	custom_steel_350_490	custom_steel_350_490	custom_steel_350_490	Yes
`material.cross_bracing`	custom_steel_350_490	custom_steel_350_490	custom_steel_350_490	Yes
`material.end_diaphragm`	custom_steel_350_490	custom_steel_350_490	custom_steel_350_490	Yes
`material.deck`	custom_concrete_40_3_5	custom_concrete_40_3_5	custom_concrete_40_3_5	Yes
`material.girder.fy`	350	350	350	Yes
`material.girder.density`	80.0	80.0	80.0	Yes
`material.deck.density`	26.0	26.0	26.0	Yes
`typical_section.deck_thickness`	250.0	250.0	250.0	Yes
`typical_section.wearing_course.density`	24.0	24.0	24.0	Yes
`design_options.shear_studs.diameter`	20	20	20	Yes
`design_options_cont.partial_factor.yielding_and_buckling.gamma_m0`	1.10	1.1	1.1	Yes

Table 6.1 is the important one for custom materials. Density 80 is in both dictionaries. The analysis still used 78500 N/m3. So the UI and the dicts are honest; _build_material_props is not reading density. That is defect 10.1.

Table 6.2 TC06 - additional inputs reach both dictionaries

Key	I entered	Input dict	Output dict	Match
`geometry.span`	30.0	30.0	30.0	Yes
`geometry.carriageway_width`	7.5	7.5	7.5	Yes
`geometry.skew_angle`	0.0	0.0	0.0	Yes
`material.girder`	custom_steel_350_490	custom_steel_350_490	custom_steel_350_490	Yes
`material.cross_bracing`	custom_steel_350_490	custom_steel_350_490	custom_steel_350_490	Yes
`material.end_diaphragm`	custom_steel_350_490	custom_steel_350_490	custom_steel_350_490	Yes
`material.deck`	custom_concrete_40_3_5	custom_concrete_40_3_5	custom_concrete_40_3_5	Yes
`material.girder.fy`	350	350	350	Yes
`material.girder.density`	78.5	78.5	78.5	Yes
`material.deck.density`	26.0	26.0	26.0	Yes
`typical_section.deck_thickness`	280.0	280.0	280.0	Yes
`typical_section.wearing_course.density`	22.0	22.0	22.0	Yes
`design_options.shear_studs.diameter`	22	22	22	Yes
`design_options_cont.partial_factor.yielding_and_buckling.gamma_m0`	1.15	1.15	1.15	Yes

On TC06 the deck thickness 280 mm, wearing-course density and stud diameter 22 mm all match. The CAD deck thickness matched 280 mm as well, so additional inputs are not being silently replaced by defaults for those fields.

Table 6.3 TC07 - three different steel grades are stored

Key	I entered	Input dict	Output dict	Match
`geometry.span`	30.0	30.0	30.0	Yes
`geometry.carriageway_width`	7.5	7.5	7.5	Yes
`geometry.skew_angle`	0.0	0.0	0.0	Yes
`material.girder`	custom_steel_350_490	custom_steel_350_490	custom_steel_350_490	Yes
`material.cross_bracing`	custom_steel_250_410	custom_steel_250_410	custom_steel_250_410	Yes
`material.end_diaphragm`	custom_steel_280_440	custom_steel_280_440	custom_steel_280_440	Yes
`material.deck`	custom_concrete_40_3_5	custom_concrete_40_3_5	custom_concrete_40_3_5	Yes
`material.girder.fy`	350	350	350	Yes
`material.girder.density`	80.0	80.0	80.0	Yes
`material.deck.density`	25.0	25.0	25.0	Yes
`typical_section.deck_thickness`	250.0	250.0	250.0	Yes
`typical_section.wearing_course.density`	24.0	24.0	24.0	Yes
`design_options.shear_studs.diameter`	20	20	20	Yes
`design_options_cont.partial_factor.yielding_and_buckling.gamma_m0`	1.10	1.1	1.1	Yes

TC07 was meant to check whether girder, cross bracing and end diaphragm can be different steels. The dictionaries do keep three grades. The design steel object is still built from the girder only (defect 10.4).

Table 6.4 TC01 - failed run: input dictionary after set_input (no complete output)

Key	I entered	Input dict	Output dict	Match
`geometry.span`	30.0	30.0	-	Input only
`geometry.carriageway_width`	7.5	7.5	-	Input only
`material.girder`	E 350A	E 350A	-	Input only
`material.deck`	M40	M40	-	Input only

TC01 never produced an output dictionary because design() raised on Gs. Geometry and the grade name did reach input_dict. The gap is the missing Gs key, not a failure to store span.

7. Steel members and the deck

The task asked to cover steel members and the deck, not only span and materials. Numbers below are from TC02 unless I say otherwise. They are the same keys the Output dock and the design-report chapters read.

Table 7.1 Utilisation on the output dock (TC02)

Output-dock key	%	Remark
util.flexure	19.51	Within 100
util.shear	39.91	Within 100
util.interaction	19.66	Within 100
util.ltb	22.81	Within 100
util.long_trans_shear	19.82	Within 100
util.fatigue	157.26	Above 100; design() still finished
util.stress_limitation	24.87	Within 100
util.deflection_crack	45.78	Within 100

Flexure, shear, LTB and deflection are all under 100%. Fatigue is 157%. The module still built CAD. I treated that as a defect (10.5), not as a pass on the check itself.

Table 7.2 Steel dock vs output dictionary (TC02)

Output key	Value	What it is
steeldesign.details.grade_of_material	custom_steel_350_490	Steel grade on the dock
steeldesign.details.section_designation	1670 × 510 × 22 × 510 × 22	Welded girder
steeldesign.details.total_depth	1670.0	Girder depth (mm)
steeldesign.details.web_thickness	10.0	Web (mm)
steeldesign.details.top_flange_width	510.0	Top flange width (mm)
steeldesign.details.shear.diameter	20.0	Shear stud (mm)
steeldesign.details.stiffener_summary.method	post_critical	Stiffener method
steeldesign.details.stiffener_summary.end_count	4	Bearing stiffeners

Table 7.3 Deck report bag (TC02)

Key	Value	What it is
deck.report.span	2.1	Girder spacing 2.1 m
deck.report.fy	500.0	Fe 500 from Design Options
deck.report.w_dl	6.1312500000000005	Slab dead load
deck.report.m_uls_sag	45.9407878432377	ULS sagging moment
deck.report.as_req_bot	596.2922946746502	Required bottom As
deck.report.punch_ok	True	Punching
deck.report.shear_ok	True	One-way shear

On TC06 the extra section data is visible in the output: deck thickness 280.0 mm (CAD 280.0 mm); wearing course density 22.0 kN/m3, thickness 80.0 mm; stud entered 22 mm, dock 22.0 mm.

On TC07 I set girder density 80.0, cross bracing 70.0, end diaphragm 75.0. The analysis density was 78500.0 (girder default).

8. Three designs in a row after Unlock

Unlock on the desktop calls backend.reset() and clears the results. I needed to know whether the next Design still drew the previous span. TC_MULTI_01 uses one PlateGirderBridge instance, calls reset(), changes span / carriageway / skew, and designs again. The case fails if the old span is still in the output dictionary or if the CAD span does not move.

Run	Planned L / CW / skew	Output L / CW / skew	CAD span (mm)	Stale?
1	25.0 / 7.5 / 0.0	25.0 / 7.5 / 0.0	25000.0	False
2	30.0 / 8.5 / 5.0	30.0 / 8.5 / 5.0	30000.0	False
3	35.0 / 9.5 / 10.0	35.0 / 9.5 / 10.0	35000.0	False

Suite result: **PASS**. Stale data found: False. Run 1 is 25 m, run 2 is 30 m, run 3 is 35 m. CAD span in millimetres is 25000, 30000, 35000. The live window in Section 9 and the video follow the same order, with Unlock in between.

9. What the desktop window actually showed

The figures in this section are full-window captures of OsdagBridge. I kept each picture inside the page margin. Look at the left dock for the numbers I typed, and at the right pane for CAD or plots after Design.

Custom materials, then Design

Figure 1 is before Design. Girder and Cross bracing are set to custom_steel. Span is 30 m, carriageway 7.5 m. Figure 2 is the same session after Design: the dock is locked and the OCC view shows the deck, girders and bracing. That is the custom-material run the task asked for.

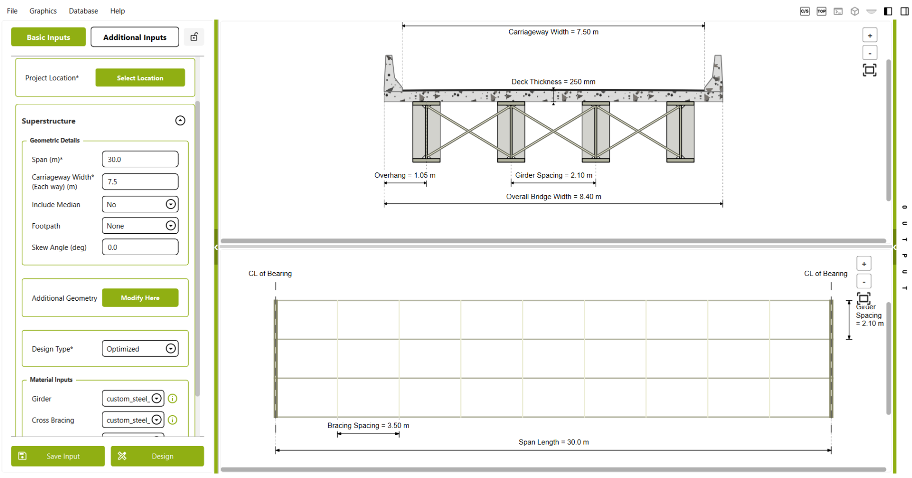

Figure 1. Before Design. Custom steel on Girder and Cross bracing; span 30 m.

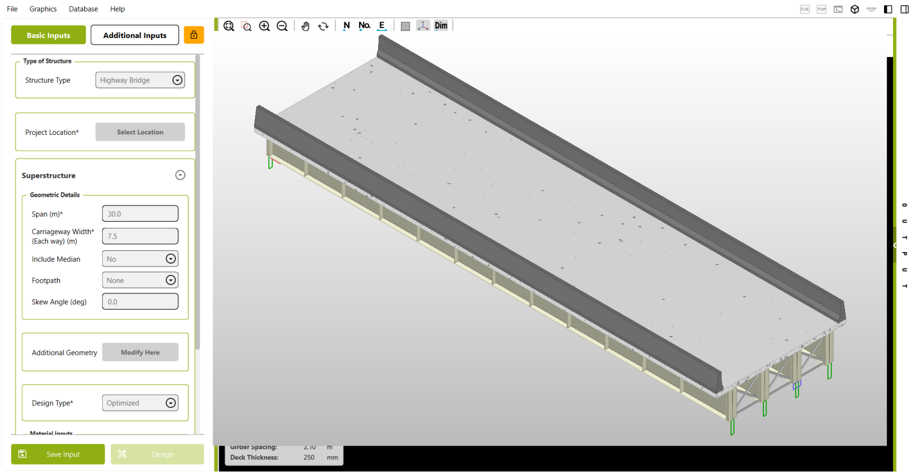

Figure 2. After Design. OCC 3D CAD of the girder bridge in the same window.

Plots dock, then Unlock

Figure 3 is the Plots dock inside the same window, not a file export. Figure 4 is Unlock: the CAD is gone and Design is available again, which is the starting point for the next run.

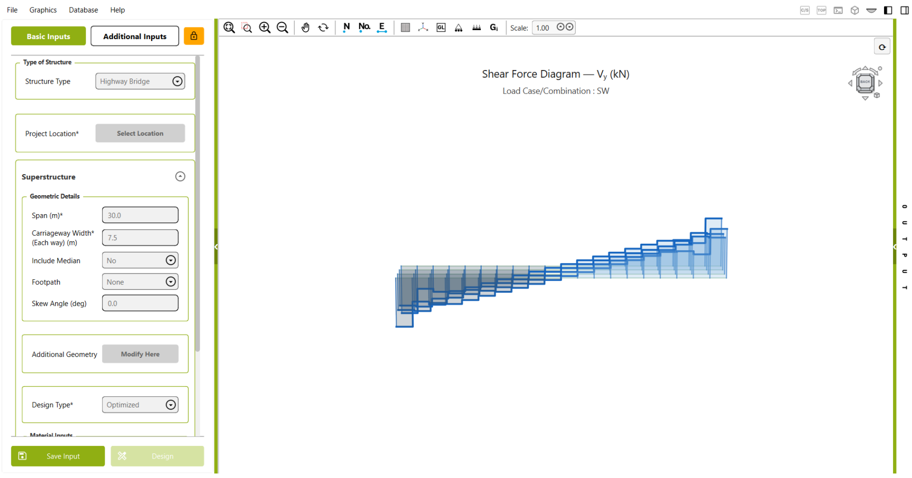

Figure 3. Plots dock. Shear-force diagram in the live window.

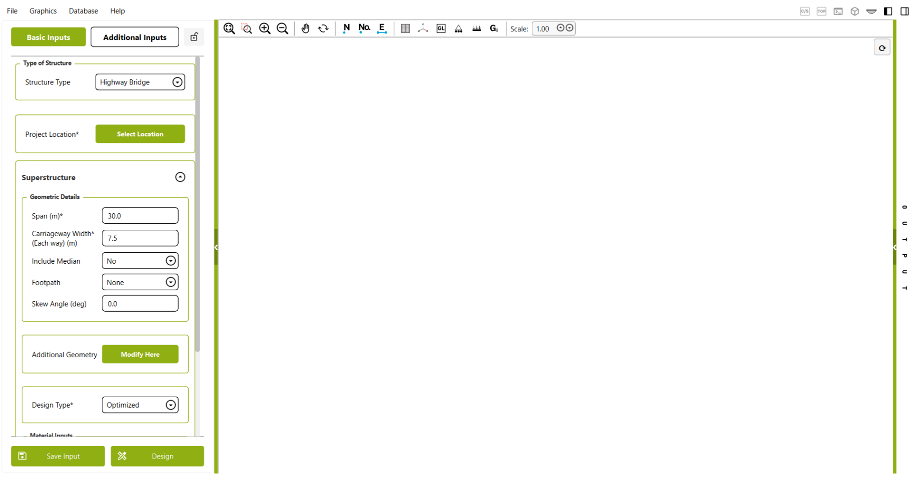

Figure 4. After Unlock. Results cleared, ready for the next design.

Second and third design

Figure 5 is the second design (span 25 m). Figure 6 is the third (span 35 m, skew 10 deg). The model on the right follows the new geometry. It is not the 30 m model left on screen.

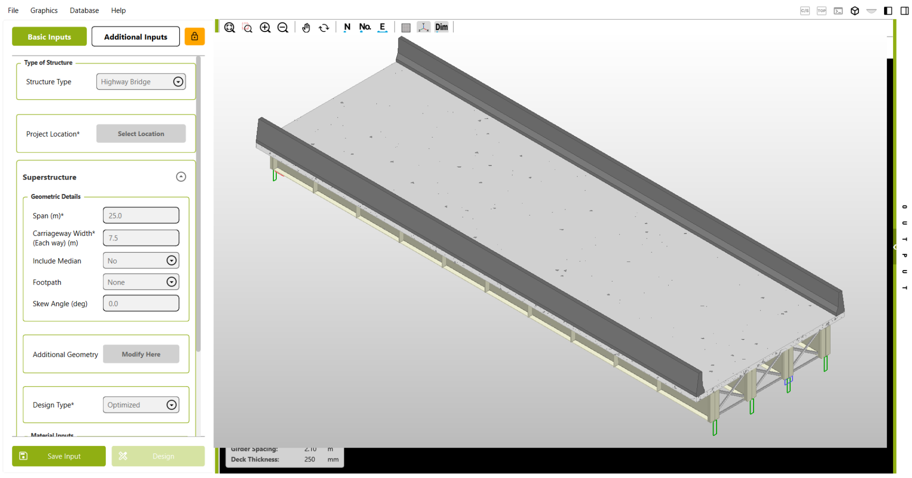

Figure 5. Second design after Unlock. Span 25 m; CAD updated.

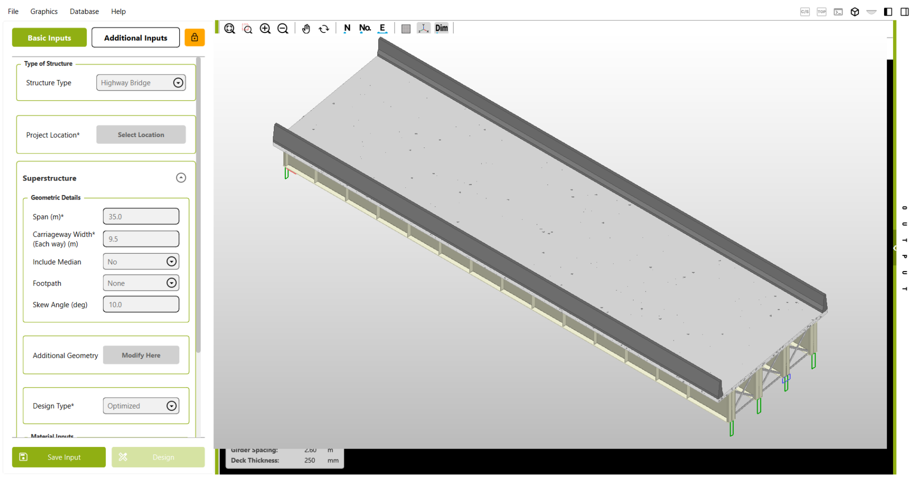

Figure 6. Third design. Span 35 m, skew 10 deg; CAD is not leftover from run 2.

Plots produced by the dock generators (TC02)

Figures 7 and 8 are the bending-moment and shear envelopes from the same functions the Plots dock calls after a custom-material design. They are here to show the dock is not empty. The window evidence is Figure 3 and the video.

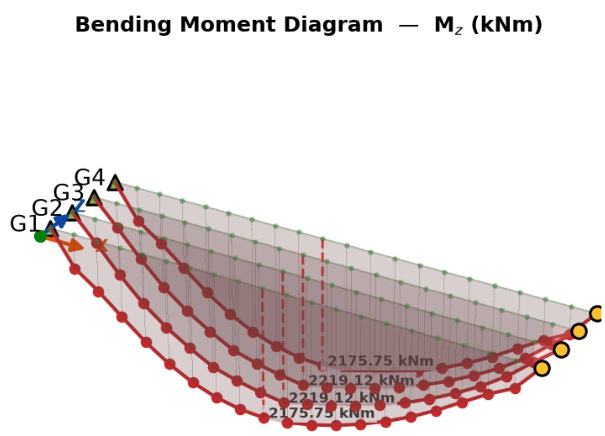

Figure 7. Bending-moment envelope (Mz), TC02.

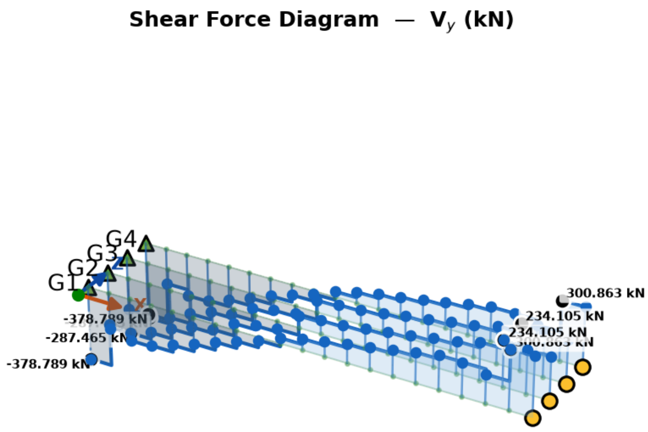

Figure 8. Shear-force envelope (Vy), TC02.

Exported OCC 3D CAD (evidence/screenshots/cad_3d)

Besides the live GUI window, I exported real OCC views with the same headless Viewer3d.ExportToImage path the product uses for report figures. Files live under evidence/screenshots/cad_3d/ (iso / front / top / end). That covers visual accuracy of the 3D CAD model for custom materials, skew/median, custom section, and the three Unlock redesigns (CAD span 25000 → 30000 → 35000 mm). Full checklist: test_cases/GRAPHICAL_OUTPUT_VERIFICATION.md.

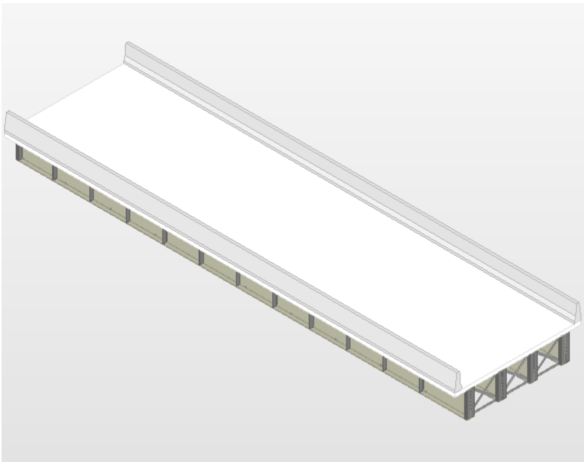

Figure 9. OCC isometric — TC02 custom materials (girders, deck, bracing, stiffeners).

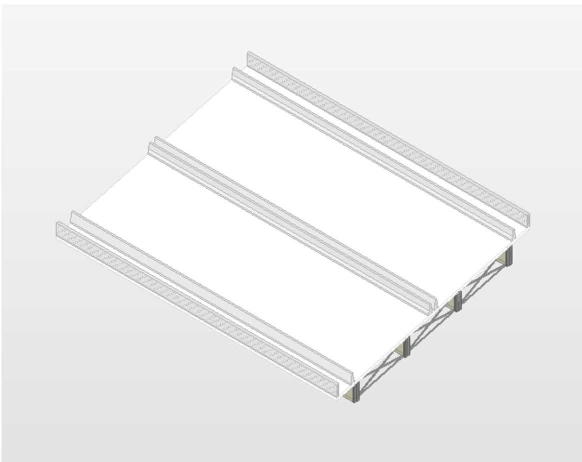

Figure 10. OCC isometric — TC03 skew + median + footpath layout.

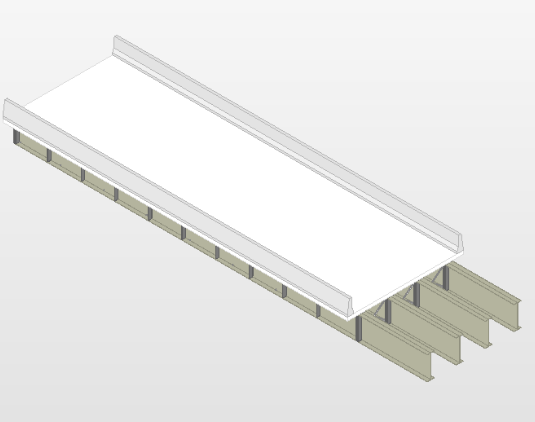

Figure 11. Multi-run 1 — CAD span 25 m after Design.

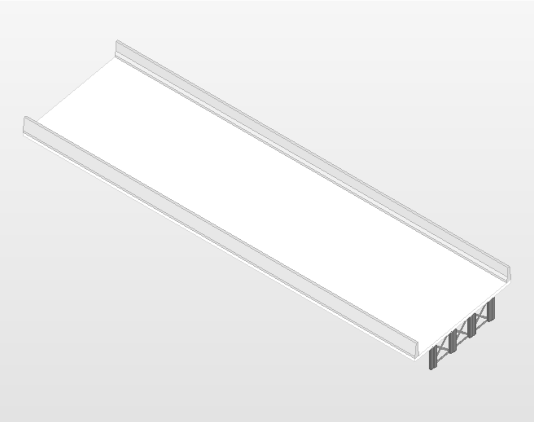

Figure 12. Multi-run 3 — CAD span 35 m (not a stale 25 m model).

10. Bugs I found, and the files that should change

I did not patch these in the clone used for the log, so the failed cases are still failed. Each item below links to a live issue on github.com/Nidhikhare12/OsdagBridge/issues. Local notes and suggested patches are also in issues/ and issues/ISSUE_LINKS.md. #19 and #20 already existed; they were reopened with fresh evidence. #41-#45 are new.

10.1 Custom steel density is ignored (#20)

File: plategirderbridge.py, _build_material_props. Custom fy, fu and E are read from input_dict. Density falls back to 78500 N/m3. TC02 stores density 80 in both dictionaries; analysis does not use it. Fix: when the database lookup is empty, read KEY_MATERIAL_GIRDER_DENSITY and convert kN/m3 to N/m3.

10.2 Grade E 350A has no Gs (#42)

File: designer.py, BridgeConfig.from_plate_girder_bridge. KEY_MATERIAL_GIRDER_G is required. TC01 and TC04 fail. TC08 completes when Gs and deck fck are primed the way the GUI dialog would. Fix: store Gs in SQLite, or set G = E / (2(1+nu)) when Gs is missing.

10.3 OSI load does not restore the custom-material map (#19)

File: input_dock.py, populate_from_dict. It does not rebuild _material_custom_fields. After a cold OSI load, _prime_material_inputs can overwrite custom fy, E and density. TC10 keeps the keys at API level. Fix: rebuild the map from loaded sub-keys, or skip priming when they already exist.

10.4 Cross-bracing and end-diaphragm custom grades are unused (#43)

One SteelProperties object is built from the girder grade only. TC07 stores distinct CB/ED fy and density; global design steel follows the girder. Fix: pass CB/ED properties into those members, or say in the UI that only girder steel is used.

10.5 Fatigue utilisation above 100% does not fail design() (#44)

TC02, TC05, TC06, TC07 finish and still emit CAD with util.fatigue between 157% and 176%. Fix: set a failed-design flag when any util.* exceeds 100, or ask before showing CAD.

10.6 Custom deck density is ignored (#41)

ConcreteProperties has no density field. add_dead_loads() calls create_deck_load with thickness only, so 25 kN/m3 is used. TC02 and TC06 enter 26. Fix: pass KEY_MATERIAL_DECK_DENSITY into create_deck_load.

10.7 Missing DCR categories are written as 0.0 (#45)

store_design_results uses category_urs.get(..., 0.0)*100. On TC03, flexure/shear/fatigue show 0.0 while LTB is 20.44%. A skipped check looks like a safe 0%. Fix: write None when the category is absent.

11. Video

The file OsdagBridge_Testing_Demo.mp4 is a silent recording of the same desktop window as Section 9. There is no title card. The whole application window is in frame (letterboxed, not cropped).

Order of shots: (1) custom steel and deck in Basic Inputs, span 30 m; (2) after Design, OCC 3D CAD; (3) Plots dock; (4) Unlock; (5) second design, span 25 m; (6) third design, span 35 m and skew 10 deg, CAD and plots updated. That covers custom materials, three consecutive runs, and CAD/plots. Upload the mp4 to YouTube (unlisted) or Drive and put the public URL in the form.

12. What I take away

Custom and additional inputs are stored from the values I typed, through input_dict, into output_dict. Three designs after Unlock do not keep a stale span or a stale CAD. The live window shows custom materials, 3D CAD and plots. Girder, stiffener, shear-connector and deck-report quantities are present after a passing design.

Default E 350A cannot finish until Gs (and deck fck) are on input_dict; TC08 isolates that. Custom steel and deck densities are stored but not applied in analysis. Cross-bracing and end-diaphragm custom grades are not consumed. Fatigue UR can exceed 100% without aborting design. Both the passing cases and the two failures are in the log, with the exact parameter sets the task asked for.